ch. 2.3.

$q = n \cdot e$, $F_e = \frac{k q q}{r^2}$, $k = \frac{1}{4\pi E_0}$

$F_g = \frac{g m m}{r^2}$, $E = \frac{F}{q}$, $E = \frac{k q}{r^2}$

rod $E = \frac{kQ}{a(a+L)}$, ring $E = \frac{kQx}{(x^2+a^2)^{3/2}}$

Disk $E = 2\pi\sigma k \left[1 - \frac{x}{(x^2+R^2)^{1/2}}\right]$, $\rho = \frac{Q}{V}$

$\sigma = \frac{Q}{A}$, $\lambda = \frac{Q}{L}$, $\frac{N_2}{N_1} = \frac{Q_2}{Q_1}$

$F = ma$, $a = \frac{qE}{m}$, $v_2 = v_1 + at$

$d_2 = d_1 + vt + \frac{1}{2} a t^2$, $v_2^2 = v_1^2 + 2ad$

$x_2 = x_1 + v_1 t$

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Ch. 2. 4.

$\Phi = \bar{E} \cdot \bar{A} \cos\theta$, $\Phi = \frac{\Sigma q_{in}}{E_0}$ ← هي العلاقة

inside sphere : $E = \frac{kQr}{a^3}$, outside sphere : $E = \frac{kQ}{r^2}$

infnite rod : $E = \frac{2k\lambda}{r}$, infnite plane : $E = \frac{\sigma}{2E_0}$

Ch. 2. 5.

$\Delta U = -q\int_A^B E \cdot dr$, $\Delta V = -\int_A^B E \cdot ds$, $V = \frac{U}{q}$

$U = -qE \cdot d$, $V = -E \cdot d$, $V = \frac{kq}{r}$, $V = k(\frac{q}{r} + \frac{q}{r} \dots)$

$U = \frac{k \cdot q \cdot q}{r}$, $U = k(\frac{qq}{r} + \frac{qq}{r})$, $E_x = \frac{-dV}{dx}$ ...

ring : $V = \frac{kQ}{\sqrt{a^2+x^2}}$, rod : $V = k\frac{Q}{L}\ln\left(\frac{L+\sqrt{L^2+a^2}}{a}\right)$

Disk : $V = k\sigma 2\pi\left[(R^2+x^2)^{\frac{1}{2}} - x\right]$

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ch. 2.6

$C = \frac{Q}{\Delta V}$, sphere: $C = 4\pi E_0 q$, spherical: $C = \frac{ab}{k(b-a)}$

parallel plates: $C = \frac{E_0 \cdot A}{d}$, cylindrical: $C = \frac{L}{2k \ln(\frac{b}{a})}$

parallel: $Q = Q_1 + Q_2 \dots$, $C_{eq} = C_1 + C_2 \dots$, $V_{eq} = V_1 = V_2 \dots$

series: $Q = Q_1 = Q_2 \dots$, $V = V_1 + V_2 \dots$, $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \dots$

$U = \frac{1}{2} Q \Delta V$, $\Delta V = \frac{V_0}{k}$, $U = \frac{U_0}{k}$, $E = \frac{E_0}{k}$

$C = C_0 k$, $Q = Q_0$

ch. 2.7.

$I = \frac{\Delta Q}{\Delta t}$, $I = n \cdot q \cdot A \cdot V$, $J = \frac{I}{A}$, $J = \sigma E$

$\rho = \frac{1}{\sigma}$, $R = \frac{\Delta V}{I}$, $R = \frac{\rho l}{A}$, $R = R_0 (1 + \alpha (T - T_0))$

$\rho = \rho_0 (1 + \alpha (T - T_0))$, $P = VI = \frac{V^2}{R} = RI^2$

$Cost = P \cdot \Delta t \cdot price$

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Ch. 2. 8

$\mathcal{E} = IR + Ir$, $\Delta V = \mathcal{E} - Ir$, $I = \frac{\mathcal{E}}{R+r}$, $\mathcal{E} = V + Ir$

$P_{\mathcal{E}} = R_R + R_r$, **series**: $I = I_1 = I_2$, $V = V_1 + V_2$, $R_{eq} = R_1 + R_2$

parallel: $I = I_1 + I_2$, $V = V_1 = V_2$, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

kirchof Rule: $\Sigma I_{in} = \Sigma I_{out}$, $\Delta V = 0$

Charging: $q = q_{max}(1 - e^{-t/\tau})$, $I = I_{max}(e^{-t/\tau})$

pischarging: $q = q_{max}(e^{-t/\tau})$, $I = -I_{max}(e^{-t/\tau})$

$q_{max} = \mathcal{E}C$, $I_{max} = \frac{\mathcal{E}}{k}$, $\tau = CR$

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